Recommendation System Project

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What is a Recommendation System?

A recommendation system is an AI algorithm primarily associated with machine learning. It uses datasets to recommend and predict what consumers would like based on given information about a consumer. This can include previous purchases, ratings, likes and dislikes. Any information the recommendation system has access to it will use it to predict and suggest relevant items to consumers.

Given Dataset

The dataset we are using for the recommendation system is from CiaoDVD, which is a dataset crawled from the DVD category of the dvd.ciao.co.uk product review website in December 2013. Specifically, we will use the dataset "movie-ratings.txt." This dataset contains a list of ratings by different users. It includes six features, those being userId, movieId, movie-categoryId, reviewId, the movie rating from 1 to 5, and the date of the review. Through our EDA, we have found these key findings:

- The dataset contains 72,665 movie ratings from 17,615 users and 16,121 movies.
- The rating matrix is extremely sparse, with a sparsity of 99.97%.
- There are no missing values in the dataset.
- There are no duplicate rows in the dataset.
- The rating distribution is skewed toward higher ratings, especially 4-star and 5-star ratings.
- Ratings per user and ratings per movie both show a long-tail pattern: most users rate only a few movies, and most movies receive only a few ratings.

Global Average

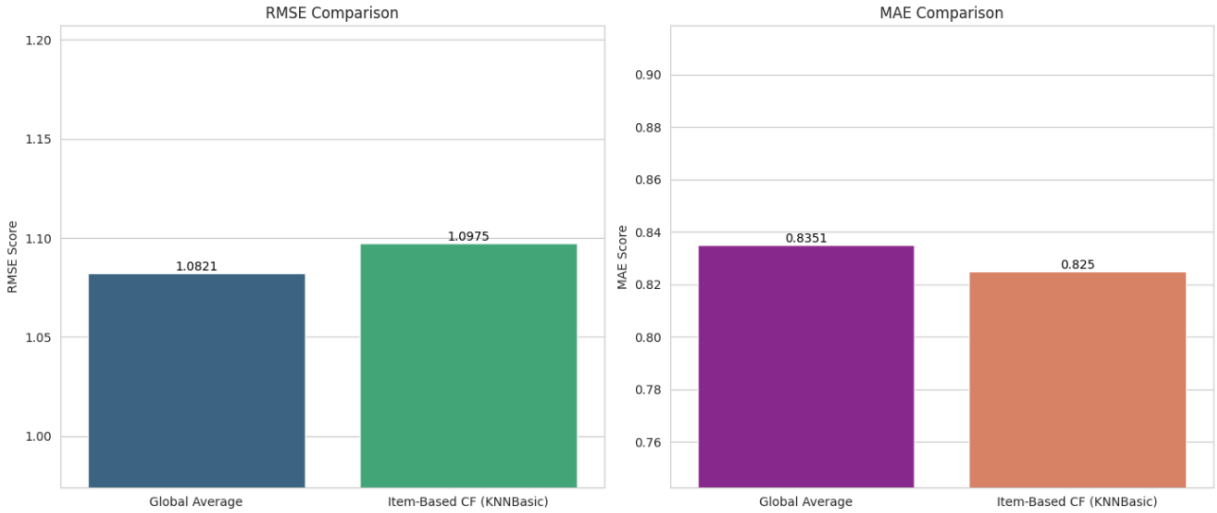
The first recommendation algorithm that was implemented was global average. This algorithm simply takes the mean of every rating in the training set, and uses that number as a prediction for any unknown value. It is a baseline predictor for other more complex algorithms to compare to.

Item-Based CF (KNN)

The other recommendation algorithm used was the item-based collaborative filtering using KNN. Unlike user-based CF, which finds similar ratings made by users, item-based CF finds items that are rated similarly. The algorithm then takes the ratings of those items(movies) from a user, and predicts the unknown ratings. While being more complex than global average, this algorithm is still relatively straightforward in the way it makes predictions.

Results

After evaluating the performance of both models using RMSE and MAE, they each ended up being relatively similar with global average having a higher MAE but a lower RMSE compared to item-based CF.



This similarity in performance could be due to the distribution of data being quite imbalanced with most ratings in the data being 4 or 5. The sparsity of the data could also be a factor with the ratings for each movie being relatively few. This sparsity of data makes it more difficult for item-based CF to look for similar items.

Extra Credit: Top-K Evaluation and Trust Network Analysis

For extra credit, we completed both optional CSIT 360 opportunities: Top-K evaluation and trust network analysis.

For Top-K evaluation, we calculated Precision@K and Recall@K using K = 5 and K = 10. We used a leave-one-positive-out setup, where one movie rated 4 or 5 was held out for each eligible user. The model then ranked that held-out movie together with 100 sampled unrated movies.

Top-K Extra Credit Results

Model	K	Precision@K	Recall@K
Global Average	5	0.0092	0.0461
Global Average	10	0.0099	0.0992
Item-Based CF	5	0.0269	0.1346
Item-Based CF	10	0.0235	0.2349

The Item-Based CF model outperformed the Global Average baseline for Top-K recommendation because it uses movie similarity based on user rating patterns.

We also analyzed the optional trust network file, trusts.txt. The trust network contains 40,133 trust links, 1,438 unique trustors, 4,299 unique trustees, and 4,658 unique users overall. The average trust rating is 1.0. There are no duplicate trust links and no self-trust links. The network density is 0.001850, showing that the trust network is very sparse.

Vibe Coding Reflection

The use of AI assistants in this project was very helpful in our research. We used it to help understand the concepts of each algorithm as well as create code snippets and help with solving errors. The main way of typing prompts was by simply asking questions in

the same way you would ask another human a question. Many AI assistants are programmed to make the assistant more human, making it easier for people to use it and try and get their questions answered. From this project, we have realized that AI and the use of AI assistants are very powerful tools in data science, which can help certain processes such as researching, writing code and bug fixing much simpler for users to make sense of.